Project Design Phase

Solution Architecture

Date	4 march 2026
Team ID	SWTID-2026-2725
Project Name	SpendWise
Maximum Marks	4 Marks

Solution Architecture

- The SpendWise application is designed with a simple backend architecture that helps beginners clearly understand how backend systems work. The architecture focuses on connecting APIs, authentication, and database operations in a structured way. The project is developed using Node.js, Express.js, and MongoDB to demonstrate modern backend development practices.
- The system follows a client–server model. In this model, the client (such as Postman, Thunder Client, or a simple frontend) sends HTTP requests to the backend server. The backend processes these requests, performs necessary operations such as authentication or data management, and communicates with the MongoDB database to store or retrieve information.

Goals of the Solution Architecture

- Identify the Best Technology Stack

- The project uses reliable and beginner-friendly technologies for backend development.
- Node.js is used as the runtime environment for executing JavaScript on the server.
- Express.js helps in creating REST APIs and managing server routes easily.
- MongoDB is used as a NoSQL database to store application data.
- Mongoose is used to create schemas and manage database interactions.
- JWT (JSON Web Token) is implemented to provide secure user authentication and protect certain routes.
- Postman or Thunder Client are used to test and verify API functionality.

Define Structure and System Characteristics

- SpendWise follows a clean and organized project structure to make the backend easy to understand and maintain.
- Controllers handle the main application logic and process user requests.
- Models define the structure of the data stored in the MongoDB database.
- Routes define the API endpoints that users interact with.
- Middleware manages tasks like verifying JWT tokens and protecting secure routes.

The architecture is designed to be:

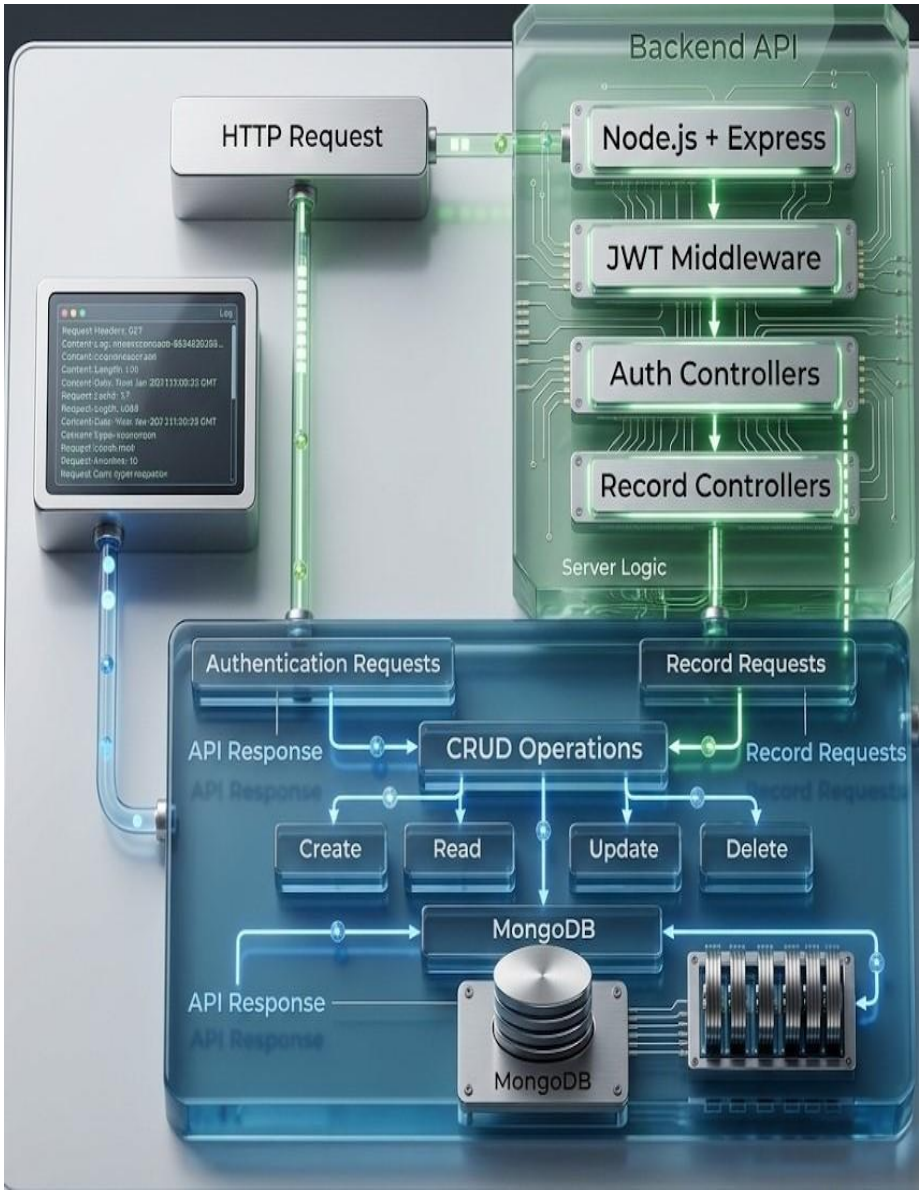
- Easy for beginners to learn backend development concepts
- Maintainable so that developers can modify or update features easily
- Scalable so additional features such as analytics or frontend integration can be added in the future

Outline Features and Development Stages

The development process of SpendWise is divided into several stages to ensure systematic implementation.

- ✓ Stage 1: Initialize the Node.js project and configure MongoDB database connection.
- ✓ Stage 2: Implement user authentication features including registration and login using JWT.
- ✓ Stage 3: Add JWT middleware to secure private API routes.
- ✓ Stage 4: Develop CRUD operations to allow users to create, view, edit, and delete their records
- ✓ Stage 5: Test and validate the APIs using Postman or Thunder Client and prepare documentation.

Define Development and Delivery Specifications

- To ensure proper functioning of the system, certain development standards are followed.
- The backend must provide RESTful API endpoints for user authentication and data management.
- Secure routes must verify the JWT token before allowing access.

- All user information and application data must be stored in MongoDB.
- The code should be written in a structured and modular format for better readability and learning.
- The application should run locally using Node.js and be easily configurable using environment variables.
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